Geometric Representation and Error Modeling of Grover's Algorithm

Russell Dillon
Advisor: Dr. Ela Celikbas

January 2026

1 Abstract

Quantum algorithms can outperform classical methods for certain tasks, and understanding them requires a precise mathematical framework. We study this framework using linear algebra and matrix groups. This project focuses on Grover's search algorithm, which finds a marked item among N possibilities in about $\sqrt{N}$ steps rather than N steps classically. We model quantum states as vectors in C^{2^n} and gates as unitary matrices, and we highlight a geometric interpretation: the oracle and diffuser act like reflections whose product acts as a rotation toward the marked state in a two-dimensional invariant subspace. We also simulate Grover's algorithm under three error models: depolarization, thermal relaxation, and readout errors, and compare computational accuracy to a noiseless search procedure.

2 Introduction

Having first been born in the late twentieth century, quantum computing has been a developing field for over 30 years. Behind the theory of quantum computing are the core devices of quantum mechanics. Linear Algebra, dominating the mathematics of quantum mechanics, and Group Theory, explaining gate structure, are invaluable tools to describe algorithms in quantum computing.

2.1 Group Theory

Group Theory is an important branch of abstract algebra that studies structures called **groups**. A **group**, denoted (G, $\circ$), refers to any set of elements **G**, and a binary operation ($\circ$) associated with these elements that satisfies four fundamental axioms:

Axiom 1: Closure Closure requires that performing your group operation between any two elements in your set produces another element in the set:

$$\forall \text{ a, b} \in \text{G, if a} \circ \text{b = c, then c} \in \text{G}$$

Axiom 2: Associativity Given any three elements of your **group**; a, b, and c; the following is true:

$$\forall \text{ a, b, c} \in \text{G, (a} \circ \text{b)} \circ \text{c = a} \circ \text{(b} \circ \text{c)}$$

Or, similarly, the order in which you perform adjacent operations does not change the result of an expression.

Axiom 3: Existence of an Identity Element For any **group** G, there exists one unique identity element, **e**, such that:

$$\forall \, g \in G, \, g \circ e = e \circ g = g$$

Or, composing any arbitrary element, g, with your identity element returns g.

Axiom 4: Existence of Inverses For any **group** G, every element g ∈ G, g has an inverse g^{-1} such that:

$$\forall \, g \in G, \, g \circ g^{-1} = g^{-1} \circ g = e$$

Where e is the identity element of the **group**.

2.1.1 Examples of Groups

A simple example of a group is the set of all integers under the operation of addition. Denoted by ($\mathbb{Z}$, +), it is easy to verify that this group satisfies all four fundamental axioms of a group.

Closure Adding any two integers together results in another integer.

Associativity It is well known that Addition is associative.

Existence of an Identity Element The identity element of $(\mathbb{Z}, +)$ is 0, as for any integer, $z \in \mathbb{Z}, \; z + 0 = 0 + z = z$

Existence of Inverses Every integer $z \in \mathbb{Z}$ has an additive inverse, $z^{-1} = -z$, which satisfies the condition $z + z^{-1} = z^{-1} + z = 0$

Another group example, relevant to the content of this paper, is the Unitary Group of $n \times n$ matrices, denoted $(U(n), \times)$. The Unitary group is the set of all matrices whose **hermitian conjugate** is also its inverse under the operation of matrix multiplication. In set builder notation, the unitary group is:

$$U(n) = \{ \; A; \; A^{-1} = A^{\dagger} \; \}$$

Where A is an $n \times n$ square matrix, and $A^\dagger$ is the **hermitian conjugate** of A. It is also easy to show that the four group axioms are satisfied under this set and operation. A fact that will be explored further is that all quantum operators must be unitary, and thus form the unitary group. It is also the case that the unitary group is infinitely differentiable, and thus forms a smooth manifold in the Euclidean space $\mathbb{R}^{N^2}$. This results in the realization that not only is the state vector of a system a geometric object, but so is the set of all valid quantum gates.

2.1.2 The Hermitian Conjugate

Given a 2×2 matrix A = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $a, b, c, and\, d \in \mathbb{C}$, the **hermitian conjugate** of A, or $A^\dagger$, is given by:

$$A^\dagger = \begin{bmatrix} a^* & c^* \\ b^* & d^* \end{bmatrix}$$

is the conjugate transpose of A.

2.2 Quantum Mechanics

Quantum mechanics is the product of over a decade's worth of work from many notable physicists, and it aims to explain the behavior of the universe on a microscopic scale. The mathematics governing quantum mechanics is also undoubtedly different from that of classical mechanics, which relies heavily on linear algebra and partial differential equations. To explain the state of a particle, or a group of particles, on a scale where they behave in accordance with quantum mechanics, we refer to the **wavefunction** of the system, $|\psi\rangle$, defined as

$$|\psi\rangle = \sum c_i |x_i\rangle$$

Where c_i is the probability amplitude corresponding to the state $|x_i\rangle$. An example of such a state is the Bell State $|\psi^+\rangle$ = $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. It is important to note that for a quantum state to be valid when performing quantum computations, the state must be normalized, or:

$$\sqrt{\sum |c_i|^2} = 1$$

A two-state wavefunction, for example, would be of the form:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

And the complex conjugate of a wavefunction $|\psi\rangle^* = \langle\psi|$, where

$$\langle\psi| = \alpha^* \langle 0| + \beta^* \langle 1|$$

Now consider the state $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$. This state is definitely normalized, as

$$(\frac{\sqrt{3}}{2})^2 + (\frac{1}{2})^2 = \frac{3}{4} + \frac{1}{4} = 1$$

Because the probability of measuring a state x_i is $|c_i|^2$, it is apparent that the normalization means that the total probability of measuring every state is one. This makes sense, as you cannot have a probability $\neq 1$ for measuring anything at all.

We can also express the wavefunction above in a more compact form. We can express our two-state wavefunction in the form

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

the conjugate of $|\psi\rangle$ can be expressed as

$$\langle\psi| = \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix}$$

Because we can express a quantum state as a column or row vector, it is the case that any n-qubit system is isomorphic to a unit vector in a 2^n-dimensional complex Euclidean space.

2.2.1 Quantum Computing

Quantum Algorithms are the quantum alternative to classical algorithms. In a classical algorithm, you would apply logical gates to bits to execute a desired task. In a quantum algorithm, a similar process is used: you apply quantum gates to qubits, manipulating their wavefunction, to achieve a desired result. The state of an individual qubit can be thought of as a point on a three-dimensional unit sphere, called the **Bloch Sphere**:

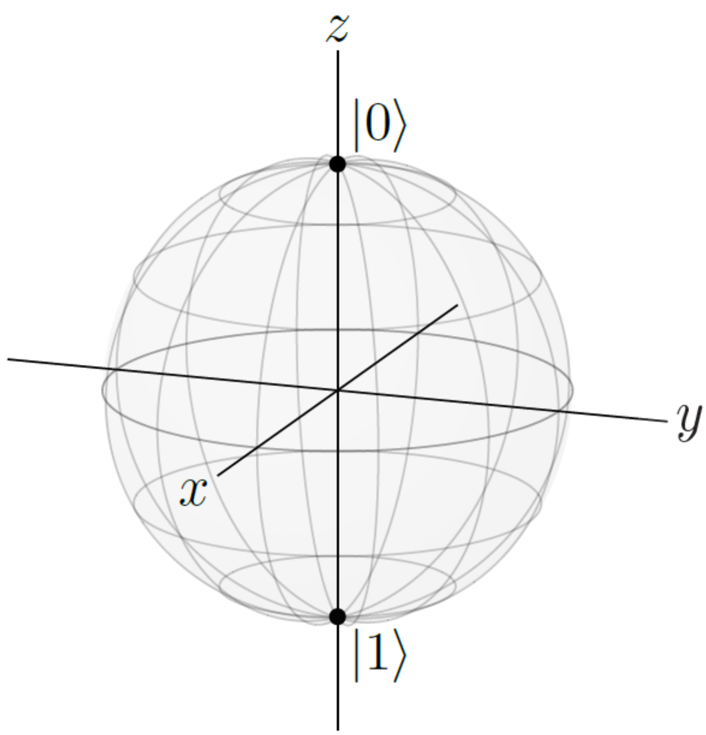


Figure 1: A diagram of the Bloch Sphere. The Bloch Sphere is useful for visualizing the superposition a single qubit is in. A state is represented by a point on this sphere, and when measurement occurs, the state vector collapses to one of the poles, depending on which basis you measure your qubit in.

In notation, the states at the ends of the Z-axis, $|0\rangle$ (”off”) and $|1\rangle$ (”on”), form the computational basis that a quantum system is measured in. The coordinates of a qubit’s state vector relative to the X and Y axes help visualize a superposition of the on and off states.

2.2.2 Circuit Complexity

While the time complexity of an algorithm is not unique to quantum computing, it is an important metric in determining the efficiency of an algorithm. Called ”Big O Notation,” O(f(N)) is considered the time-complexity of an algorithm, where f(N) represents the average number of operations a computer must perform given N inputs. For example, the binary search algorithm has a time complexity of O($\log N$). If you have to search through a sorted list of N items, your computer, on average, performs $\log N$ operations. Note that this is more efficient than Grover's Algorithm, and is classical, but requires that the list you are searching through is sorted beforehand. Under optimal sorting conditions, the best sorting algorithms can achieve a time complexity of O(N) (these take longer under sub-optimal conditions). Combining the best Sorting and Searching Algorithms, you achieve a time complexity of O($N \log N$), which is still slower than Grover's.

2.3 The Pauli and Clifford Groups

The Pauli and Clifford groups contain quantum operators that are integral to quantum computing theory. The Pauli group, denoted *P* is:

$$P = \{\text{I, X, Y, Z}\}$$

Where the I gate is the identity operator, and the X, Y, and Z operators represent $180°$ single-qubit state vector rotations around these respective axes on the Bloch Sphere.

The group of Clifford gates, *C*, is defined to be the set of all matrices that map a Pauli matrix to another Pauli matrix under conjugation, or:

$$C = \{ \text{U}: UPU^\dagger \in \text{P}\}$$

Some quantum operators contained within the Clifford group are the Hadamard gate (H), the Controlled-NOT gate (CNOT), and the Phase gate (S).

2.4 Quantum Gates as Rotations and Reflections

Because a valid quantum computing gate is unitary with complex entries, any quantum operator acting on a state vector must be norm-preserving. If a unitary operator of dimension N acts on a state vector, $v \in \mathbb{C}^N$, as follows

$$|U_N(v)| = |v|$$

Then the unitary operator, U, must either be a rotation or reflection in $\mathbb{C}^N$. This has interesting implications that will be explored later.

3 Grover's Algorithm

In 1996, an Indian-American computer scientist, Lov Grover, developed a quantum computing algorithm that fundamentally changed the field. Grover's Algorithm provides optimal search in an unsorted database and offers a quadratic speedup over the best classical brute-force search algorithms.

3.1 The Searching Problem

Consider having a large database of linked variables you have to search through, like medical records. Databases like these may link information such as the names, addresses, or phone numbers of patients. Consider a database like so:

Name	Address	Phone Number
Alice	308 Negra Arroyo Lane	505-503-4455
Bob	42 Wallaby Way	805-457-4992
Charlie	1600 Pennsylvania Ave	914-737-9938
David	221B Baker Street	618-625-8313

Now, you may notice that this list is sorted in alphabetical order (with respect to the names), and consequently, searching for a name would actually be quite efficient using a classical binary search algorithm. In fact, binary searching has a time complexity of O($log(n)$). If you wanted to search for "Alice" in the database above, it would take about two queries to find the right name.

Instead, however, if you wanted to search for an address, you have two (non-quantum) options: First, you can sort your database, then search efficiently classically. This would have a time complexity of O($n * log(n)^2$), combining the time complexities of the sorting and searching algorithms. For small databases, this is a manageable task for a classical computer, but it becomes costly for the large databases seen in modern infrastructure. The second option is to search item-by-item until you find the address you were looking for. If, for example, you wanted to look for "221B Baker Street," it would take four queries (searching top-to-bottom). It is less efficient to sort a database, then search it, compared to guessing-and-checking if the database contains more than 10 items.

Grover's Algorithm makes searching for any variable more computationally efficient for unsorted databases of size $n = 5$ or larger. This is a significant speedup over searching in the classical regime.

3.2 The Phase Oracle

Recall that a wavefunction, $|\psi\rangle$ can be expressed as $|\psi\rangle = \sum c_i |x_i\rangle$, where c_i is the probability amplitude corresponding with the quantum state $|x_i\rangle$. If we wanted to modify a specific state, or set of states, such that they have an overall phase change of -1, we would need an operator that multiplies a target set of states by -1, and leaves every other state unchanged. This operator is known as the Phase Oracle, U_f. The phase oracle contains a binary function, $f(x)$, such that U_f operates on a wavefunction $|\psi\rangle$ like:

$$U_f \circ |\psi\rangle = U_f \circ \sum c_i |x_i\rangle = \sum -1^{f(x)} c_i |x_i\rangle$$

Because the function f is binary and only outputs 0's or 1's, we can mark a target state, $|w\rangle$, by crafting a function, f, such that $f(w) = 1$ and $f(x) = 0$ when $x \neq w$. To characterize the Phase oracle as a matrix, we can use the fact that a quantum operator acting on a wavefunction is represented by left matrix multiplication. Consider the two-state case when there is only one qubit in a quantum circuit being acted on by the phase oracle:

$$U_f \circ |\psi\rangle = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a * \alpha + b * \beta \\ c * \alpha + d * \beta \end{bmatrix}$$

Because U_f is only supposed to change the phase of a state by a factor of ± 1, it must be the case that U_f is a strictly diagonal matrix with entries ± 1, depending on which state you wish to apply a phase to.

$$U_f = \begin{bmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{bmatrix}$$

Consider the case when we want to apply a negative phase to the $|0\rangle$ state. In this case, our phase oracle would have to be

$$U_f = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

so that

$$U_f \circ |\psi\rangle = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -\alpha \\ \beta \end{bmatrix} = -\alpha |0\rangle + \beta |1\rangle$$

In fact, modifying the n'th state in a wavefunction requires the n'th term along the main diagonal of the phase oracle to be changed. Generalizing this definition for larger n is trivial.

3.2.1 Showing U_f is a Quantum Operator

For the phase oracle to be a valid quantum operator, it must be represented by a unitary matrix, or by definition

$$U^\dagger = U^{-1}$$

Because the phase oracle is a strictly diagonal matrix with real entries, it follows that

$$U_f^\dagger = U_f$$

Therefore, in the case of the phase oracle, we must show that

$$U_f^\dagger = U_f = U_f^{-1}$$

We know that the phase oracle of arbitrary lengths looks like

$$U_f = \begin{bmatrix} \pm 1 & 0 & 0 & \cdots \\ 0 & \pm 1 & 0 & \cdots \\ 0 & 0 & \pm 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Therefore

$$U_f \circ U_f = \begin{bmatrix} (\pm 1)^2 & 0 & 0 & \cdots \\ 0 & (\pm 1)^2 & 0 & \cdots \\ 0 & 0 & (\pm 1)^2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} = I$$

And it follows that $U_f = U_f^\dagger = U_f^{-1}$. This means that the phase oracle is a valid quantum operator.

3.3 The Diffusion Operator

The diffusion operator is a reflection of $|\psi\rangle$ about the uniform superposition state, $|s\rangle = \frac{1}{\sqrt{2}} \sum |+\rangle$. For a two-state system, this is defined as:

$$R_s = 2\,|\psi\rangle\,\langle\psi| - I$$

The diffusion operator is important in many algorithms and, in Grover's algorithm, is used in combination with the phase oracle to amplify the probability amplitude corresponding to the search target.

To show R_s is a valid quantum operator , like with the phase oracle, we must show that:

$$R_s^\dagger = R_s^{-1}$$

Using our definition for R_s, we can show that the hermitian conjugate of the diffusion operator is the same as the diffusion operator:

$$\begin{aligned} R_s^\dagger &= (2\left|\psi\right\rangle\left\langle\psi\right| - I)^\dagger \\ &= (2\left|\psi\right\rangle\left\langle\psi\right|)^\dagger - I^\dagger \\ &= 2\left\langle\psi\right|^\dagger\left|\psi\right\rangle^\dagger - I \\ &= 2\left|\psi\right\rangle\left\langle\psi\right| - I \\ &= R_s \end{aligned}$$

Like with the phase oracle, it now suffices to show that $R_s \circ R_s = I$ for the diffusion operator to be unitary:

$$\begin{aligned} R_s \circ R_s &= (2\left|\psi\right\rangle\left\langle\psi\right| - I) \circ (2\left|\psi\right\rangle\left\langle\psi\right| - I) \\ &= (2\left|\psi\right\rangle\left\langle\psi\right|)^2 - 4\left|\psi\right\rangle\left\langle\psi\right| + I \\ &= 2\left|\psi\right\rangle\left\langle\psi\right| 2\left|\psi\right\rangle\left\langle\psi\right| - 4\left|\psi\right\rangle\left\langle\psi\right| + I \\ &= 4\left|\psi\right\rangle\left\langle\psi\right|\left|\psi\right\rangle\left\langle\psi\right| - 4\left|\psi\right\rangle\left\langle\psi\right| + I \\ &= 4\left|\psi\right\rangle\left\langle\psi\right| - 4\left|\psi\right\rangle\left\langle\psi\right| + I \\ &= I \end{aligned}$$

and $R_s^\dagger = R_s^{-1}$, meaning the diffusion operator is unitary.

3.4 Grover's Algorithm as Reflections and Rotations

The circuit composition of Grover's Algorithm is relatively simple. You first begin by placing all qubits in the $\left|+\right\rangle$ state via a Hadamard gate, then alternate applying the phase oracle and diffusion operator until your system's wavefunction aligns with your search target. The circuit is:

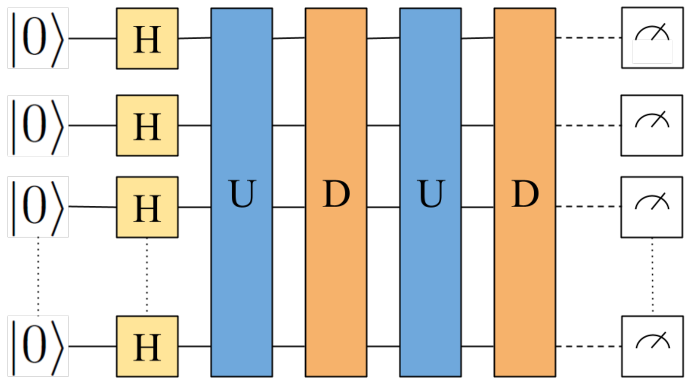


Figure 2: A circuit diagram of Grover's Algorithm. The circuit starts with Hadamard (H) gates applied to all qubits, followed by the phase oracle and diffusion operator.

Mathematically, you start with all qubits in the off state:

$$|\psi\rangle = |000...00\rangle$$

After applying Hadamard gates to each qubit, your wavefunction is:

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |i\rangle$$

where i is the binary form of all numbers between 0 and N. Here, we can observe that when searching for a bit string, which we will denote as w, there is only one unique target state. We can separate w from all other states as:

$$|\psi\rangle = \frac{1}{\sqrt{N}} |w\rangle + \sqrt{\frac{N-1}{N}} \sum_{i \neq w} |i\rangle$$

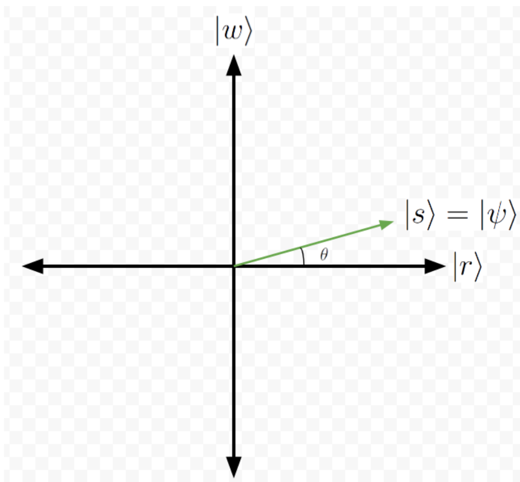
To simplify this further, we can recognize that all bit strings are actually orthogonal vectors that form a computational basis. We can then group all states that are not our target state together as the pseudo-state $|r\rangle$, for "rest."

$$|\psi\rangle = \frac{1}{\sqrt{N}} |w\rangle + \sqrt{\frac{N-1}{N}} |r\rangle$$

Another qualitative observation is that both probability amplitudes are real-valued and less than one. We can liken them to a point on a unit sphere in a two-dimensional coordinate system, and rewrite our wavefunction as

$$|\psi\rangle = \sin\theta\,|w\rangle + \cos\theta\,|r\rangle\,;\ \sin\theta = \frac{1}{\sqrt{N}},\ \cos\theta = \sqrt{\frac{N-1}{N}}$$

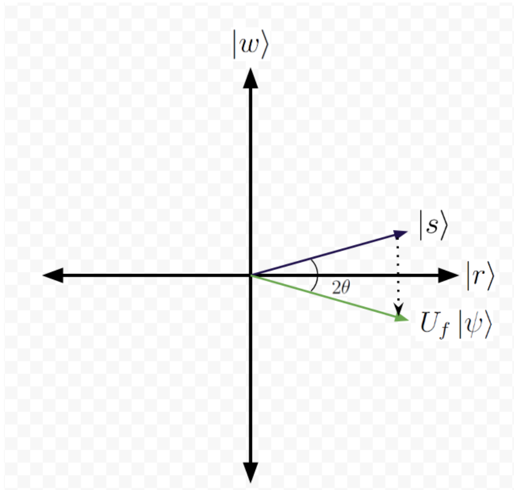
Visualizing this, we can draw our coordinate system with the y-axis being labeled $|w\rangle$ and our x-axis being $|r\rangle$



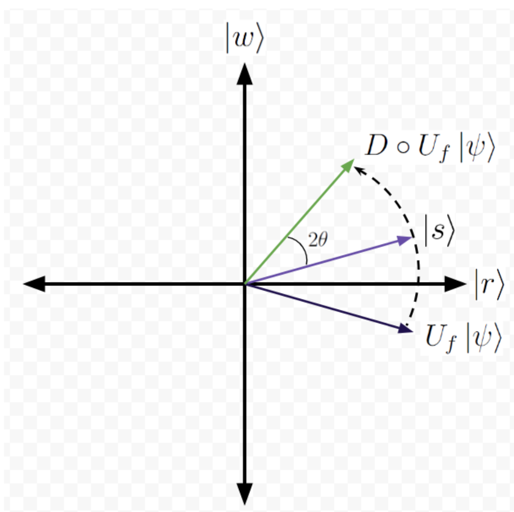
Next, we apply the phase oracle to our system, and our wavefunction becomes

$$|\psi\rangle = -\sin\theta\,|w\rangle + \cos\theta\,|r\rangle$$

flipping the y-coordinate of our state vector:



Finally, we apply our diffusion operator. Writing out mathematically what happens to our wavefunction does not provide sufficient insight into the geometric view of Grover's Algorithm, so I will forego that step and show you our two-dimensional coordinate system after reflecting $|\psi\rangle$ about the uniform superposition state, $|s\rangle$:

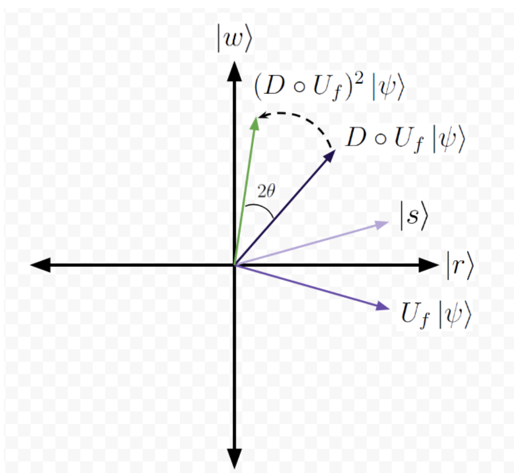


And we see that the combination of the phase oracle and diffusion operator is simply a reflection in our computational state space. You repeat this process until your state vector rotates as close to the target state as possible, and then you collapse your wavefunction, which has a significant probability of returning the target state.

Even though this interpretation of Grover's shows these rotations in a lower-dimensional state space, these operations are actually acting on a vector space, V $\in \mathbb{C}^N \cong \mathbb{R}^{2N}$

3.5 Time Complexity of Grover's Algorithm

The composition of the phase oracle and the diffusion operator is known as a single "Grover Operation." To determine how many Grover Operations we need to provide optimal measurement of the target state, we look at the diagram of our two-dimensional state space after applying the diffuser:



We need to rotate our state vector $\frac{\pi}{2} - \theta$ radians to place it on the $|w\rangle$ axis. Because every Grover Operation rotates our state vector by 2θ, the total rotation ($\frac{\pi}{2}$) can be expressed in terms of t, the number of times we have to apply our Grover Operation:

$$\frac{\pi}{2} \approx t(2\theta) + \theta \rightarrow t \approx \frac{\pi}{4\theta} - \frac{1}{2}$$

We know that $\frac{1}{\sqrt{N}} = \sin\theta \rightarrow \theta = \sin^{-1}(\frac{1}{\sqrt{N}})$. For large N (where classical searching struggles, $|\frac{1}{\sqrt{N}}| \ll 1$, and we can use the approximation that $\sin^{-1}(\frac{1}{\sqrt{N}}) \approx \frac{1}{\sqrt{N}}$:

$$t \approx \frac{\pi}{4\frac{1}{\sqrt{N}}} - \frac{1}{2}$$
$$\approx \frac{\pi}{4}\sqrt{N}$$

and in ideal conditions, it takes $\frac{\pi}{4}\sqrt{N}$ Grover Operations to successfully execute Grover's Algorithm. This indicates that the time complexity of Grover's Algorithm is O($\sqrt{N}$) and the speedup over classical searching is quadratic. It has been proven that quantum computers can not perform any brute-force algorithms faster than O($\sqrt{N}$) [5]. The proof follows from the logic that when you separate your target state from all other possible measurement outcomes, you can only "eliminate" options (via state vector rotation) at a rate proportional to $\frac{1}{\sqrt{N}}$.

3.6 Generalizing to All Quantum Operators

This image of the phase oracle and diffuser operating on a complex vector space can be generalized to all quantum operators. As all quantum operators are unitary operators that act on quantum state vectors, they can all be interpreted as reflections. It is also true that the complex vector space $\mathbb{C}^N$ is isomorphic to the real vector space $\mathbb{R}^{2N}$, and we can interpret all quantum operators as reflections in a real vector space of dimension 2N.

4 Simulation of Grover's Algorithm

It is possible to simulate Grover's Algorithm on classical computers. We have achieved this simulation for variable numbers of qubits, and modeled Grover's with three error models, comparing accuracy rates to those of ideal simulations. The error models chosen were thermal relaxation, depolarization, and readout error.

4.1 Quantum Error

Error in quantum computing algorithms, often referred to as quantum noise, is the process of quantum information in your state vector being lost unintentionally.

This information loss can happen due to varying factors, and the type of information loss can vary depending on the physical implementation of your qubits. The general form of a two-state wavefunction is

$$|\psi\rangle = \sin\frac{\theta}{2}|0\rangle + e^{i\phi}\cos\frac{\theta}{2}|1\rangle$$

where θ is the angle of the state vector relative to the x-y plane and ϕ is the azimuthal angle in the x-y plane, relative to the positive x-axis. The kind of information that can be lost in the execution of an algorithm consists of the "wandering" of the angles θ and ϕ, bit-flipping, and measurement probability changing. Thermal relaxation is a form of bit-flip error, and depolarization is a form of losing information stored in the probability amplitudes of states.

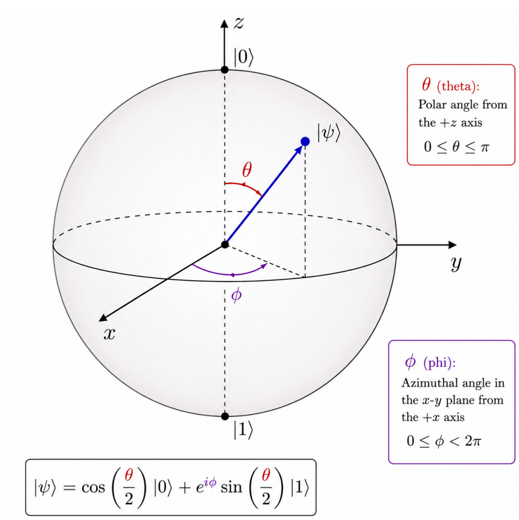


Figure 3: A diagram of the Bloch Sphere with θ and ϕ labeled. With phase errors, these angles can begin to wander or flip entirely.

Thermal relaxation error occurs when you have a qubit that is in an excited state ($|1\rangle$), that decays to the ground state ($|0\rangle$). This decay typically occurs after a qubit remains in the excited state for an amount of time related to its energy and the environment's temperature. This fact makes it important for algorithms to execute with minimal circuit depth if the goal is to minimize thermal relaxation. The primary reason why this error was chosen for simulation was due to the difficulty in correcting this error, making it a present obstacle. Semiconductor qubits are some of the most popular prospective qubits and are the most prone to thermal relaxation.

Depolarization error, the loss of phase information, occurs when a qubit's state vector tends towards the maximally mixed state. For a single qubit, the point at the center of the Bloch Sphere represents the maximally mixed state. When this happens, all information regarding phase and polarity is lost, and measuring in any of the three bases results in a purely random result. Of course, this is an extreme case, as even if the state vector only moves slightly, it is still a depolarization error. Depolarization error is prevalent in all types of qubits, as interactions with the environment will cause a loss of information regardless of your physical qubit of choice.

Readout Error is a less punishing form of error, as it is modeled by a slim probability of flipping a bit at the end of an algorithm. Compared to depolarization and thermal relaxation error, which can occur at any time during the algorithm's execution, readout error is a cause of your method of collapsing your wavefunction and measuring your system. If your measurement is too aggressive, it may change the quantum information stored in your state vector. However, comparing this relatively weak error model to others that are more severe helps give perspective on why it is important to develop error-correcting protocols.

The ideal case is when we assume that all errors are corrected (or don't happen at all!). This is a raw statistical evaluation of the algorithm, and a good measurement of how close we can rotate the state vector towards the target state. Simulating the ideal case provides a good baseline for what real implementations of Grover's Algorithm should strive to achieve.

4.2 Simulation Procedure

To simulate Grover's algorithm, we utilized Google Colab, a cloud-based Jupyter Notebook. The primary tool for the simulation of quantum algorithms is called Qiskit. Qiskit is a software development kit (SDK) developed by IBM, one of the leading companies in quantum computing, for interfacing with quantum computers.

We simulated this search procedure on five qubits (or 32 items). While this simulation is able to run with more than five qubits, we chose five to decrease the algorithm run time and clutter in the results graphics. To compare our error models versus an ideal simulation, we ran the circuit 1,000,000 (100,000 shots $\times$ 10 trials) times for each error model and plotted a histogram of the results. For readability of results, we made our target state the bit string of the binary number $|11111\rangle$. After collecting data, we calculated and compared the accuracy of all four models.

4.3 Results

For each of our error models, we ran the circuit 100,000 (one hundred thousand) times, ten times, to achieve a total of 1,000,000 shots for each model. Below is a histogram of measurement counts comparing the four models:

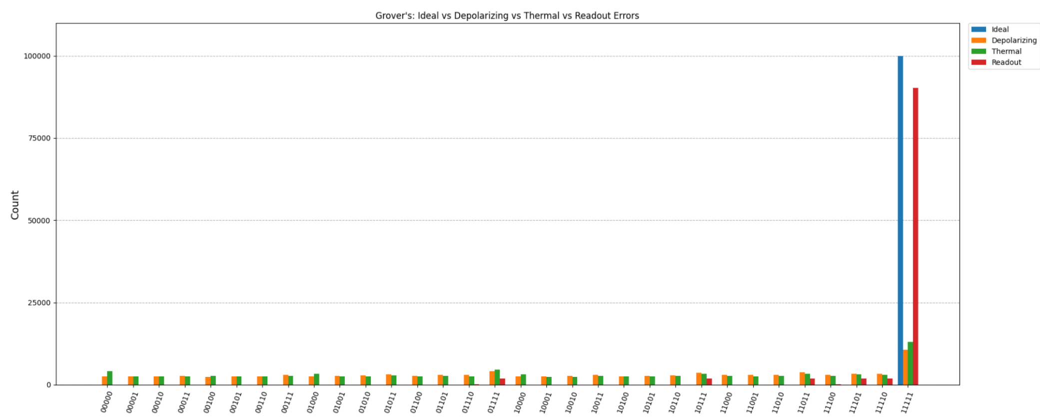


Figure 4: A histogram comparing a trial of all four simulation models. Readout error produced the largest accuracy of all non-ideal simulations. Significantly lower were the thermal relaxation and depolarization models. Recall, the target state is the last state, located on the far right of this histogram.

It is apparent that, with enough measurements, it would be possible to distinguish between your target state and incorrect measurements. After converting count rates to overall accuracy, we achieved the following results:

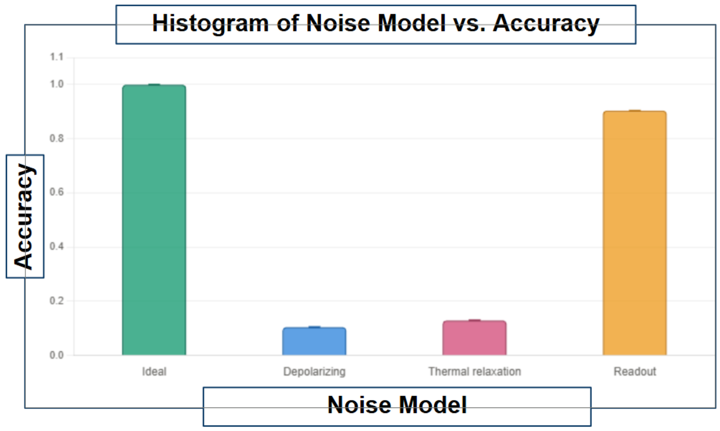


Figure 5: A histogram of the total accuracies of all models. The ideal simulation had 99.9% accuracy, readout error had 90.3% accuracy. Thermal relaxation and depolarization had accuracies of 12.9% and 10.4%, respectively.

However, as the qubit number and subsequently circuit depth increase, it would become increasingly unlikely that you measure your target state. Readout error will only scale with the number of qubits in your system, as it is only possible to acquire readout error once per qubit, at the end of your circuit. Between thermal and depolarization error, thermal error models provided the highest accuracy. In practice, depolarization models may yield slightly larger accuracy if your target bit string is not comprised of all ones. This is because an otherwise incorrect qubit state could decay from an excited, incorrect state to the correct ground state.

Circuit depth and error rates are directly correlated. Quantum noise, like thermal relaxation, is more likely to occur the longer a qubit must stay in an excited state. This means that as you have more qubits, you require more Grover Operations and time to complete the algorithm, and accuracy suffers. Errors that cause changes in the phase of a qubit, perhaps from degraded gate fidelity (accuracy of a gate) or from environmental factors, are more likely to stem from increased operations performed on qubits.

5 Conclusion

Our analysis of Grover's Algorithm provides valuable insight into the geometric intuition one can develop for quantum operators on state vectors. The abstraction that carries from the phase oracle and diffusion operator to all quantum operators is one of great interest and may provide valuable tools for evaluating other algorithms and theorizing ways to achieve other goals with quantum computing. Furthermore, our simulations of Grover's show that depolarization error is very threatening to the search procedure. This generally makes sense, as the nature of Grover's relies on the ability to distinguish between different quantum states.

Taken together, the geometric interpretation and noise models tell a powerful story. Quantum Algorithms are fundamentally geometrical objects. This fact can be exploited to analyze different sources of error and predict an algorithm's sensitivity to any given error model. In the case of Grover's Algorithm, depolarizing error produces the most damaging effect, as the algorithm relies on phase changes to converge upon the target state. Any noise that disrupts the vector space that a wavefunction lives in, like depolarizing or thermal relaxation, proves to be more dangerous to quantum algorithms than others, like readout error. This perspective generalizes well to other algorithms and could provide insight into which error models may pose a larger threat.

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